

THE TAYLOR EXPANSIONS OF $\tan x$ AND $\sec x$

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ABSTRACT. The report clarifies the relationships among the completely ordered leveled binary trees, the coefficients of the Taylor expansion of $\tan x + \sec x$ and the Pascal-like triangle with the Boustrophedon transform. The alternating permutation play the most important role in establishing these relationships.

1. INTRODUCTION

Our investigation arose from the observation that formula derived to count the number of the completely ordered leveled binary trees or COL binary trees can also be used to calculate the coefficients of the Taylor expansion of $\tan x$.

$$\tan x = x + 2\frac{x^3}{3!} + 16\frac{x^5}{5!} + 272\frac{x^7}{7!} + \dots$$

1.1. The COL binary trees.

A COL binary tree is a binary tree with the two following properties:

- Different vertices are at different levels.
- The level of each junction vertex is less than the levels of the 2 leaves.

For example, a COL binary tree with 3 leaves can be described as below:

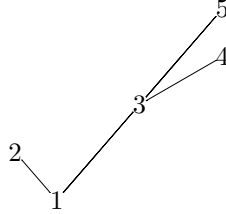


FIGURE 1

The above tree has 5 vertices. Each vertex occupies a level. The level of each junction vertex (1 or 3) is less than the levels of both terminal vertices.

The recursive formula which counts the number of COL binary trees with n leaves $T(n)$ is given by:

$$T(n) = \sum_{m=1}^{n-1} \binom{2n-2}{2m-1} T(m)T(n-m)$$

(this formula will be derived in the next part)

Taking $T(1) = 1$, we can then calculate further values of $T(n)$ for a given number of leaves $n > 1$ by using the recursive formula. The following values for $n = 2, 3, 4, 5, 6$ are as follows: $T(2) = 2, T(3) = 16, T(4) = 272, T(5) = 7936, T(6) =$

353792. These six values tend to suggest that the tree formula produces the coefficients for the Taylor expansion of $\tan x$.

1.2. The alternating permutations.

A down-up permutation $a_1 a_2 \cdots a_{i-1} a_i a_{i+1} \cdots a_n$ is an arrangement of $a_1 a_2 \cdots a_{i-1} a_i a_{i+1} \cdots a_n$ such that the elements alternatively fall and rise (the permutation starts with a fall). For example, a down-up permutation of $\{1, 2, 3, 4, 5\}$ is $2\ 1\ 5\ 3\ 4$ since the numbers start with a fall and then alternatively rise and fall. Similar definition holds for an up-down permutation except that the permutation starts with a rise.

As the COL binary tree in Figure 1 can describe the down-up permutation $2\ 1\ 5\ 3\ 4$, there is a similarity between the definition of a COL binary tree and of a down-up permutation. In other words, the number of COL binary trees with n leaves is possibly the number of down-up permutations on a set of $2n + 1$ elements. For example, there are 16 down-up permutations on a set of 5 elements (*see Appendix A*). Notice that $T(3) = 16$ is also the number of COL binary trees with 3 leaves.

1.3. The Pascal-like triangle with Boustrophedon transform.

The Boustrophedon transform applied to the sequence $\{1, 0, 0, \dots\}$ can be described as follows:

- The entry at the first row is 1 and entries at the beginning of other rows are 0.
- Every subsequent entry is the sum of the previous entry in the same row and the entry above it in the previous row [Millar, Sloane & Young, 2000].

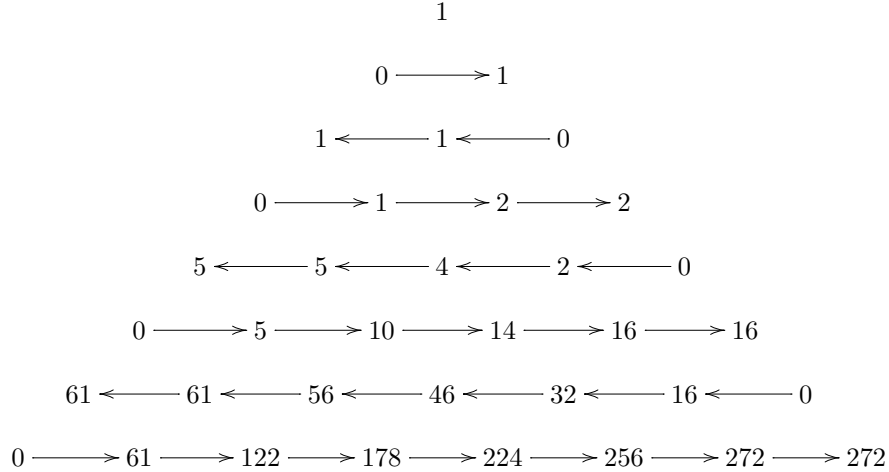


FIGURE 2

It is interesting to observe that the entries at the right side of the triangle appear to be the coefficients of the Taylor expansion for $\tan x$.

All the above observations tend to link the COL binary trees, the coefficients of the Taylor expansion of $\tan x$, the Boustrophedon transform and the alternating permutations together. Therefore, the report will examine sequentially the relationship between each of the first three components and the alternating permutations in order to establish a connection among these four components. Additionally, due

to the symmetry of $\tan x$ and $\sec x$, similar investigation about the Taylor series of $\sec x$ will also be presented. The final part of the report will discuss to some extent the relationship between the Taylor series of $\tan x$ and the Bernoulli numbers.

In this report, sections 1, 2, 3, 4 are joint work of Ryan Crompton and Pham Thi Minh Tam. Sections 5, 6 are Tam's work.

2. THE COL BINARY TREES AND THE ALTERNATING PERMUTATIONS

As can be seen from the name, a COL binary tree comes from the set $\{1, 2, 3, \dots, 2n - 1\}$ with the first level occupied by the root and having two distinguishing properties. The first property is uniqueness, that repetition or exclusion of any integers from the set is not allowed.

The second property results from the way a COL binary tree is represented by a finite sequence of numbers. Tracing around the top of the tree of n leaves beginning at the left-most leaf, we can generate a sequence of vertex level $a_1, a_2, \dots, a_{2n-1}$. The sequence is written in the form $a_1 a_2 \dots a_{2n-1}$ where each entry a_{2i-1} in the sequence is a leaf at level a_{2i-1} and each entry a_{2i} in the sequence is an internal vertex at level a_{2i} . The second property requires the vertex levels to satisfy $a_{2i} < a_{2i-1}$ and $a_{2i} < a_{2i+1}$ for $(i = 1, 2, \dots, n - 1)$.

Example: $n = 3$ leaves $\Rightarrow 2n - 1 = 5$ levels. One such tree 2 1 5 3 4 has been described in Figure 1.

Let $T(n)$ be the number of COL binary trees with n leaves. then $T(n)$ satisfies the recursive formula:

$$T(n) = \sum_{m=1}^{n-1} \binom{2n-2}{2m-1} T(m) T(n-m)$$

with the initial condition $T(1) = 1$.

Proof. Consider a COL binary tree with n leaves. By definition it will have $2n - 1$ levels and the first level is occupied by the root (the principal vertex). The proof is based on dividing the tree into the left and right components. If the left-hand component contains m leaves ($0 < m < n$), then it will contain $2m - 1$ levels. Given that the root occupies one level, then there are $\binom{2n-2}{2m-1}$ ways of choosing the $2m - 1$ levels from the remaining $2n - 2$ levels. Hence we can form $\binom{2n-2}{2m-1} T(m)$ different structures on the left-hand component. Similarly, the right-hand component has $n - m$ leaves and consequently has $T(n - m)$ structures on it. Therefore given m leaves to the left of the root, there are $\binom{2n-2}{2m-1} T(m) T(n - m)$ COL binary trees. The desired result is obtained by considering the different values for m from 1 to $n - 1$ and then summing over these values.

In order to show that the sequence generated by a COL binary tree forms a down-up permutation, consider a tree with $n + 1$ leaves. This tree forms from left to right n pairs of leaves. Since each pair requires an internal vertex, n internal vertices are needed. Hence the tree has $(n + 1) + n = 2n + 1$ vertices. Tracing these vertices generates $2n + 1$ elements of a permutation. This is a down-up permutation since the level of an internal vertex between 2 leaves is less than the levels of both.

From the permutation $a_1 a_2 \dots a_{i-1} a_i a_{i+1} \dots a_{2n+1}$, we can build a tree with $n + 1$ leaves as follows (assuming a_i is the root where $i = 2, 4, 6, 8, \dots, 2n$):

Draw a leaf at level a_1 from the root, a_2 forms an internal vertex off the branch connecting the leaf a_1 to the root and at the end of this branch is the leaf at level a_3 . Continue this procedure until the root a_i is reached at which time the right-hand

component will begin construction. This tree structure can now be seen to be in the COL binary tree form.

Hence there is a bijection between the number of COL binary trees with $n + 1$ leaves and the number of down-up permutations on a set of $2n + 1$ elements. $\square$

3. THE TAYLOR SERIES OF $\tan x$ AND $\sec x$ AND THE ALTERNATING PERMUTATIONS

3.1. The Taylor series of $\tan x$ and the alternating permutations. Consider the relationship between the coefficients of the Taylor series of $\tan x$ and the number of down-up permutations on a set with an odd number of elements.

Firstly, we will find a recursive formula for the number of down-up permutations on a set with an odd number of elements.

Let Z_{2k+1} denotes the number of down-up permutations on the set $\{1, 2, \dots, 2k+1\}$. Since 1 is the smallest element in the set, it has to be at an even position in a down-up permutation. Suppose that it is at position $2i$ for $i = 1, 2, \dots, k$.

Position	1	2	$\dots$	$2i-1$	$2i$	$2i+1$	$\dots$	$2k+1$
Number	*	*	$\dots$	*	1	*	$\dots$	*
	$\underbrace{\hspace{10em}}_{(2i-1) \text{ elements}}$					$\underbrace{\hspace{10em}}_{(2k-2i+1) \text{ elements}}$		

Since 1 is at position $2i$ in the permutation of $2k+1$ elements, there are $2i-1$ elements that precede 1 and $2k-2i+1$ elements that follow 1.

Of the remaining $2k$ elements given 1 is fixed, there are $\binom{2k}{2i-1}$ ways to choose a set of $2i-1$ elements for the left-hand permutations. Since $2i-1$ is odd, we have Z_{2i-1} down-up permutations for each set. The remaining elements also form a set of $2k-2i+1$ elements. Since this set also contains an odd number of elements, there are $Z_{2k-2i+1}$ down-up permutations for the right-hand set. Hence if 1 is at position $2i$ of a permutation length $2k+1$, the number of down-up permutations is

$$\binom{2k}{2i-1} Z_{2i-1} Z_{2k-2i+1}$$

Since 1 can be at positions $2, 4, \dots, 2k$, the number of down-up permutations on a set of $2k+1$ elements is

$$Z_{2k+1} = \sum_{i=1}^k \binom{2k}{2i-1} Z_{2i-1} Z_{2k-2i+1}$$

Now we can notice the similarity between our recursion formula derived from down-up permutations and the recursion formula derived from Joyce's COL binary trees by substituting $n = k+1$ and $i = m$. Since the number of down-up permutations on a set of 1 element is 1, we have the same initial value for both formulas. The difference arises as a result of Joyce using n as the number of leaves not the number of vertices (leaves and internal vertices) of the tree.

Next we can verify our recursion formula derived from down-up permutations by using the derivatives.

The Taylor series of $\tan x$ is:

$$f_1(x) = \tan x = \sum_{k=0}^{\infty} Z_{2k+1} \frac{x^{2k+1}}{(2k+1)!}$$

Differentiating the Taylor series for $\tan x$:

$$f_1'(x) = \sum_{k=0}^{\infty} Z_{2k+1} \frac{x^{2k}}{(2k)!} = Z_1 \frac{x^0}{0!} + \sum_{k=1}^{\infty} Z_{2k+1} \frac{x^{2k}}{(2k)!} = 1 + \sum_{k=1}^{\infty} Z_{2k+1} \frac{x^{2k}}{(2k)!}$$

We also have:

$$\begin{aligned} \tan^2 x &= \left(\sum_{n=0}^{\infty} Z_{2n+1} \frac{x^{2n+1}}{(2n+1)!} \right) \left(\sum_{m=0}^{\infty} Z_{2m+1} \frac{x^{2m+1}}{(2m+1)!} \right) \\ &= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} Z_{2n+1} Z_{2m+1} \frac{x^{2n+2m+2}}{(2n+1)!(2m+1)!} \end{aligned}$$

Let $k = m + n + 1 \Rightarrow k \geq 1$ and $0 \leq n \leq k - 1$. We have:

$$\begin{aligned} \tan^2 x &= \sum_{k=1}^{\infty} \sum_{n=0}^{k-1} Z_{2n+1} Z_{2k-2n-1} \frac{x^{2k}}{(2n+1)!(2k-2n-1)!} \\ &= \sum_{k=1}^{\infty} \sum_{n=0}^{k-1} Z_{2n+1} Z_{2k-2n-1} \frac{(2k)!}{(2n+1)!(2k-2n-1)!} \frac{x^{2k}}{(2k)!} \\ &= \sum_{k=1}^{\infty} \sum_{n=0}^{k-1} \binom{2k}{2n+1} Z_{2n+1} Z_{2k-2n-1} \frac{x^{2k}}{(2k)!} \\ &= \sum_{k=1}^{\infty} \sum_{n=1}^k \binom{2k}{2n-1} Z_{2n-1} Z_{2k-2n+1} \frac{x^{2k}}{(2k)!} \end{aligned}$$

Since $\tan' x = \tan^2 x + 1$, equating the two Taylor series, we get:

$$Z_{2k+1} = \sum_{n=1}^k \binom{2k}{2n-1} Z_{2n-1} Z_{2k-2n+1}$$

The initial value Z_1 : $f_1(x) = \tan x \Rightarrow f_1'(x) = \sec^2 x \Rightarrow f_1'(0) = 1$.

Since the number of down-up permutations on a set of one element is 1, we have the same initial value Z_1 for both formulas. Hence the coefficients of the Taylor series for $\tan x$ are given both by the recursion formula for the number of down-up permutations on a set with an odd number of elements, and by Joyce's recursion formula for counting COL binary trees with a fixed number of leaves.

3.2. The Taylor series of $\sec x$ and the alternating permutations.

A similar relationship can be obtained for the coefficients of the Taylor series for $\sec x$ and the number of down-up permutations on a set with an even number of elements. Again, we will first find a recursive formula for the number of down-up permutations on a set with an even number of elements..

Let Y_{2k} denotes the number of down-up permutations on the set $\{1, 2, \dots, 2k\}$. Since 1 is the smallest element in the set, it has to be at an even position in a down-up permutation. Again suppose that it is at position $2i$ for $i = 1, 2, \dots, k$.

Position	1	2	$\dots$	$2i-1$	$2i$	$2i+1$	$\dots$	$2k$
Number	*	*	$\dots$	*	1	*	$\dots$	*
	$\underbrace{\hspace{10em}}_{(2i-1) \text{ elements}}$				$\underbrace{\hspace{10em}}_{(2k-2i) \text{ elements}}$			

Since 1 is at position $2i$ in the permutation of $2k$ elements, there are $2i-1$ elements that precede 1 and $2k-2i$ elements that follow 1.

There are $\binom{2k-1}{2i-1}$ ways to choose a set of $2i-1$ elements for the left-hand permutations. Since $2i-1$ is odd, we have Z_{2i-1} down-up permutations for each set. The

remaining elements also form a set of $2k - 2i$ elements. Since this set contains an even number of elements, there are Y_{2k-2i} down-up permutations for the right-hand set. Hence if 1 is at position $2i$ of a permutation length $2k$, the number of down-up permutations is:

$$\binom{2k-1}{2i-1} Z_{2i-1} Y_{2k-2i}$$

Since 1 can be at positions $2, 4, \dots, 2k$, the number of down-up permutations on a set of $2k$ elements is:

$$Y_{2k} = \sum_{i=1}^k \binom{2k-1}{2i-1} Z_{2i-1} Y_{2k-2i}$$

Since the number of down-up permutations on a set of 0 or 1 element is 1, the initial conditions for this formula are $Y_0 = 1$ and $Z_1 = 1$.

Next we can verify our recursion formula derived from down-up permutations by using the derivatives.

The Taylor series of $\sec x$ is

$$f_2(x) = \sec x = \sum_{k=0}^{\infty} Y_{2k} \frac{x^{2k}}{(2k)!}$$

Differentiating the Taylor series for $\sec x$:

$$f_2'(x) = \sum_{k=1}^{\infty} Y_{2k} \frac{x^{2k-1}}{(2k-1)!}$$

We also have:

$$\begin{aligned} \tan x \sec x &= \left(\sum_{n=0}^{\infty} Z_{2n+1} \frac{x^{2n+1}}{(2n+1)!} \right) \left(\sum_{m=0}^{\infty} Y_{2m} \frac{x^{2m}}{(2m)!} \right) \\ &= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} Z_{2n+1} Y_{2m} \frac{x^{2n+2m+1}}{(2n+1)!(2m)!} \end{aligned}$$

Let $k = m + n + 1 \Rightarrow k \geq 1$ and $0 \leq n \leq k - 1$. We have:

$$\begin{aligned} \tan x \sec x &= \sum_{k=1}^{\infty} \sum_{n=0}^{k-1} Z_{2n+1} Y_{2k-2n-2} \frac{x^{2k-1}}{(2n+1)!(2k-2n-2)!} \\ &= \sum_{k=1}^{\infty} \sum_{n=0}^{k-1} Z_{2n+1} Y_{2k-2n-2} \frac{(2k-1)!}{(2n+1)!(2k-2n-2)!} \frac{x^{2k-1}}{(2k-1)!} \\ &= \sum_{k=1}^{\infty} \sum_{n=0}^{k-1} \binom{2k-1}{2n+1} Z_{2n+1} Y_{2k-2n-2} \frac{x^{2k-1}}{(2k-1)!} \\ &= \sum_{k=1}^{\infty} \sum_{n=1}^k \binom{2k-1}{2n-1} Z_{2n-1} Y_{2k-2n} \frac{x^{2k-1}}{(2k-1)!} \end{aligned}$$

Since $\sec' x = \sec x \tan x$, equating the two Taylor series, we get:

$$Y_{2k} = \sum_{n=1}^k \binom{2k-1}{2n-1} Z_{2n-1} Y_{2k-2n}$$

The initial value Y_0 and Z_1 :

$$\begin{aligned} f_2(x) &= \sec x \Rightarrow f_2(0) = 1 \\ f_1(x) &= \tan x \Rightarrow f_1'(x) = \sec^2 x \Rightarrow f_1'(0) = 1 \end{aligned}$$

We also have the same initial values Y_0 and Z_1 for both formulas. Hence the coefficients of the Taylor series for $\sec x$ are the number of down-up permutations on a set with an even number of elements.

Note: This recursive formula is the recursive formula for the expansion of $\sec x$, and also illustrate the dependency of the $\sec x$ recursion on $\tan x$. (*This independent recursive formula for $\sec x$ will be obtained later in this report but in view of the exponential generating function, not in view of the alternating permutations*).

4. THE BOUSTROPHEDON TRANSFORM

Given the sequence $a = \{a_0, a_1, a_2, \dots\}$, the boustrophedon transform (or "ox-plowing") is defined by the triangle:

$$\begin{array}{ccccccc}
 & & a_0 = b_0 & & & & \\
 & \swarrow & \dots & \searrow & & & \\
 & a_1 & \longrightarrow & b_1 = a_0 + a_1 & & & \\
 & \swarrow & & \searrow & & & \\
 b_2 = a_1 + a_2 + b_1 & \longleftarrow & a_2 + b_1 & \longleftarrow & a_2 & & \\
 & \swarrow & & \searrow & & & \\
 a_3 & \longrightarrow & a_3 + b_2 & \longrightarrow & a_2 + a_3 + b_1 + b_2 & \longrightarrow & b_3 = 2a_2 + a_3 + b_1 + b_2
 \end{array}$$

FIGURE 3

The entries at each row begin with consecutive elements of the input sequence 'a' and every subsequent entry is the sum of the previous entry in the same row and the entry above it in the previous row [Millar, Sloane & Young, 2000].

Apply the boustrophedon transform on the sequence $a = \{1, 0, 0, 0, \dots\}$, we get the triangle in Figure 2. Each entry in this triangle is given by:

$$(1) \quad \begin{array}{lll} X_{11} = 1, & X_{2k2k} = 0, & X_{2k+11} = 0 \\ X_{2kn} = X_{2kn+1} + X_{2k-1n}, & X_{2k+1n} = X_{2k+1n-1} + X_{2kn-1} \end{array}$$

Each entry down the left side of the triangle forms a coefficient of the Taylor expansion of $\sec x$. Similarly, each entry down the right side of the triangle excluding a_0 forms a coefficient of the Taylor expansion of $\tan x$. Hence non-zero entries on the left are called secant numbers and those on the right are called tangent numbers. The combination of these two sequences illustrated by the entries at the ends of the rows forms the sequence of the coefficients for the Taylor expansion of $\sec x + \tan x$.

$$\begin{aligned}
 \sec x &= 1 + 1 \frac{x^2}{2!} + 5 \frac{x^4}{4!} + 61 \frac{x^6}{6!} + \dots \\
 \tan x &= x + 2 \frac{x^3}{3!} + 16 \frac{x^5}{5!} + 272 \frac{x^7}{7!} + \dots \\
 \sec x + \tan x &= 1 + x + 1 \frac{x^2}{2!} + 2 \frac{x^3}{3!} + 5 \frac{x^4}{4!} + 16 \frac{x^5}{5!} + 61 \frac{x^6}{6!} + 272 \frac{x^7}{7!} + \dots
 \end{aligned}$$

Let X_n denote the number appearing at the end of row n (n starts from 0), then the transform has the exponential generating function:

$$\varepsilon(x) = \sum_{n=0}^{\infty} X_n \frac{x^n}{n!} = \sec x + \tan x$$

$$\begin{array}{ccccccc}
& & & & X_{11} & & \\
& & & & & & \\
& & & & X_{22} & \longrightarrow & X_{21} \\
& & & & & & \\
& & & & X_{33} & \longleftarrow & X_{32} \longleftarrow X_{31} \\
& & & & & & \\
& & & & X_{44} & \longrightarrow & X_{43} \longrightarrow X_{42} \longrightarrow X_{41} \\
& & & & \dots & \dots & \dots \dots \dots \dots \\
& & & & & & \\
& & & & X_{n-1\ n-1} & \ll & X_{n-1\ n-2} \longleftarrow \dots \longleftarrow X_{n-1\ 2} \longleftarrow X_{n-1\ 1} \\
& & & & & & \\
& & & & X_{nn} & \longrightarrow & X_{nn-1} \longrightarrow \dots \dots \dots \dots \longrightarrow X_{n2} \longrightarrow X_{n1}
\end{array}$$

FIGURE 4

This exponential generating function is the solution to the equation $\varepsilon(x)\varepsilon'(x) = \varepsilon''(x)$, given that $X_0 = X_1 = 1$. This differential equation comes from the observation that the boustrophedon transform of the X_n sequence shifted one place to the left is the same sequence shifted two places to the left [Millar et al., 2000].

Given the exponential generating function, it is easy to verify the differential equation $\varepsilon(x)\varepsilon'(x) = \varepsilon''(x)$ since

$$\begin{aligned}
\varepsilon(x) &= \sec x + \tan x \\
\varepsilon'(x) &= \sec x \tan x + \sec^2 x \\
\varepsilon''(x) &= \sec x \tan^2 x + \sec^3 x + 2 \sec^2 x \tan x
\end{aligned}$$

This differential equation suggests a way to derive a recursive formula for the coefficients of the Taylor expansion of $\sec x + \tan x$.

$$\begin{aligned}
\varepsilon(x) &= \sec x + \tan x = \sum_{m=0}^{\infty} X_m \frac{x^m}{m!} \\
\varepsilon'(x) &= \sum_{m=1}^{\infty} X_m \frac{x^{m-1}}{(m-1)!} = \sum_{n=0}^{\infty} X_{n+1} \frac{x^n}{n!} \\
\varepsilon''(x) &= \sum_{m=1}^{\infty} X_{m+1} \frac{x^{m-1}}{(m-1)!} = \sum_{k=0}^{\infty} X_{k+2} \frac{x^k}{k!} \\
\varepsilon(x)\varepsilon'(x) &= \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} X_m X_{n+1} \frac{x^{m+n}}{m!n!}
\end{aligned}$$

Let $k = m + n \Rightarrow n \leq k$, then we have:

$$\begin{aligned}
\varepsilon(x)\varepsilon'(x) &= \sum_{k=0}^{\infty} \sum_{n=0}^k X_{k-n} X_{n+1} \frac{x^k}{(k-n)!n!} \\
&= \sum_{k=0}^{\infty} \sum_{n=0}^k X_{k-n} X_{n+1} \frac{k!}{(k-n)!n!} \frac{x^k}{k!} \\
&= \sum_{k=0}^{\infty} \sum_{n=0}^k \binom{k}{n} X_{k-n} X_{n+1} \frac{x^k}{k!}
\end{aligned}$$

Therefore, by equating the two formulas, we get:

$$X_{k+2} = \sum_{n=0}^k X_{n+1} X_{k-n}$$

This is the recursion formula for $\sec x + \tan x$. The recursive formulas for $\sec x$ and for $\tan x$ can also be obtained by multiplying two Taylor series (Schwatt, 1962):

$$\begin{aligned}
\sum_{n=0}^k (-1)^n \binom{2k}{2n} E_n &= 0 \text{ with } E_n = X_{2n} \\
\sum_{n=0}^k (-1)^n \binom{2k+1}{2n+1} T_n &= 1 \text{ with } T_n = X_{2n+1} \\
&\text{(see Appendix B for the proof)}
\end{aligned}$$

5. THE PASCAL-LIKE TRIANGLE AND THE ALTERNATING PERMUTATIONS

Let E_{2k+1}^n represents the number of down-up permutations on $\{1, 2, \dots, 2k+1\}$ where the first element in these permutations is n . An example of row 5 is explained in the Appendix A. (2)

Let E_{2k}^n represents the number of up-down permutations on $\{1, 2, \dots, 2k\}$ where the first element in these permutations is n . (3)

Each entry in the triangle can be interpreted in terms of down-up permutations as follows:

- Each entry at odd rows $X_{k\ n}$ (k odd) is interpreted in terms of down-up permutations E_k^n . For example, $X_{5\ 2} = E_5^2 = 2$ since there are 2 down-up permutations on $\{1, 2, 3, 4, 5\}$ which begin with 2: $2\ 1\ 4\ 3\ 5$ and $2\ 1\ 5\ 3\ 4$.
- Each entry at even rows $X_{k\ n}$ (k even) is interpreted in terms of up-down permutations E_k^n . For example, $X_{4\ 2} = E_4^2 = 2$ since there are 2 up-down permutations on $\{1, 2, 3, 4\}$ which begin with 2: $2\ 3\ 1\ 4$ and $2\ 4\ 1\ 3$.

The first property indicates that E_{2k+1}^{2k+1} represents the number of down-up permutations on the set $\{1, 2, \dots, 2k+1\}$ beginning with $2k+1$. Since the first place is fixed, the remaining elements form up-down permutation on $\{1, 2, \dots, 2k\}$. In other words, entries at the left ends of odd rows are the numbers of up-down permutations on sets with even number of elements.

Similarly, the second property indicates that E_{2k}^1 represents the number of up-down permutations on the set $\{1, 2, \dots, 2k\}$ beginning with 1. Since the first place is fixed, the remaining elements form up-down permutation on $\{2, 3, \dots, 2k\}$. It is apparent that the number of down-up permutations on this set is also the number of down-up permutations on $\{1, 2, \dots, 2k-1\}$. In other words, entries at the right ends of even rows are the numbers of down-up permutations on sets with odd number of elements.

The triangle of the boustrophedon transform can be constructed by the following rules, which are a minor modification of (1):

$$X_{k+1\ n} = \begin{cases} \sum_{i=n}^k X_{k\ i}, & \text{when } k+1 \text{ even} \\ \sum_{i=1}^{n-1} X_{k\ i}, & \text{when } k+1 \text{ odd} \end{cases}$$

At even row: Each entry is the sum of the elements at its left-hand side in the previous row, for example, $14 = 5 + 5 + 4$. (4)

At odd row: Each entry is the sum of the elements at its right-hand side in the previous row, for example, $56 = 10 + 14 + 16 + 16$. (5)

The two above properties (2) and (3) can be proved by induction with the following hypothesis:

The triangle $\{X_{k\ n}\}$ is given by $X_{k\ n} = E_k^n$

Proof. $\{a_1, a_2, \dots, a_k\}$ is a set of integers. Let $\# \{a_1, a_2, \dots, a_k\}_{a_n}$ denote the number of down-up (up-down) permutations on $\{a_1, a_2, \dots, a_k\}$ starting with a_n when k is odd (even).

Case $n = 1$:

There is 1 down-up permutation on the set $\{1\}$ and it starts with 1, thus $E_1^1 = 1$. Hence $X_{1\ 1} = E_1^1 = 1$.

Case $n = 2$:

There is 0 up-down permutation starting with 2 on the set $\{1, 2\}$, thus $E_2^2 = 0$. Hence $X_{2\ 2} = E_2^2 = 0$.

There is 1 up-down permutation starting with 1 on the set $\{1, 2\}$, which is $1\ 2$, thus $E_2^1 = 1$. Hence $X_{2\ 1} = E_2^1 = 1$.

Case $n = 3$:

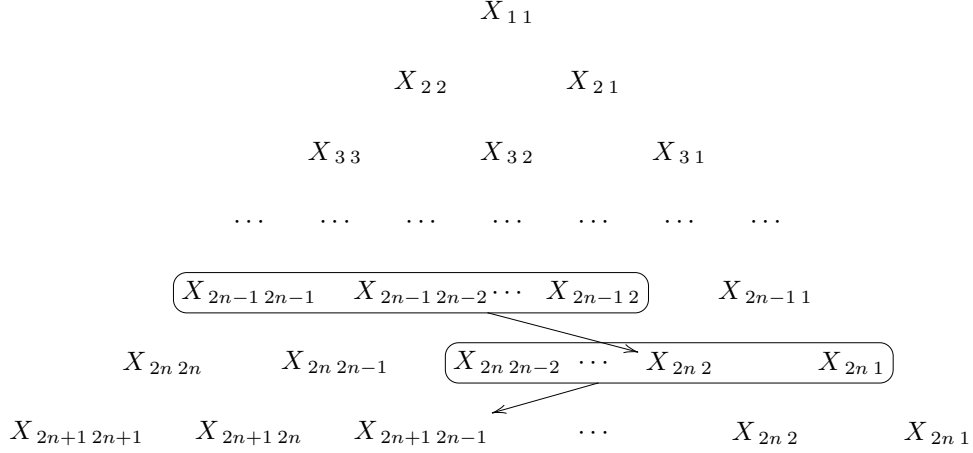


FIGURE 5

There is 1 down-up permutation starting with 3 on the set $\{1, 2, 3\}$, which is $3\ 1\ 2$, thus $E_3^3 = 0$. Hence $X_{3,3} = E_3^3 = 1$.

There is 1 down-up permutation starting with 2 on the set $\{1, 2, 3\}$, which is $2\ 1\ 3$, thus $E_3^2 = 1$. Hence $X_{3,2} = E_3^2 = 1$.

There is 0 down-up permutation starting with 1 on the set $\{1, 2, 3\}$, thus $E_3^1 = 1$. Hence $X_{3,1} = E_3^1 = 0$.

Suppose that $X_{k,n} = E_k^n$ holds for k odd which means that we have constructed a triangle with k rows satisfying $X_{m,n} = E_m^n$ for $m \leq k$, k odd given.

We will prove that $X_{k+1,n} = E_{k+1}^n$ and then that $X_{k+2,n} = E_{k+2}^n$

Firstly, we will show that $X_{k+1,n} = E_{k+1}^n$.

Since $k+1$ even, by definition (2), $E_{k+1}^n = \#\{1, 2, \dots, k+1\}_n$ is the number of up-down permutations on $\{1, 2, \dots, k+1\}$ starting with n . Since the first place is fixed with n , E_{k+1}^n is also the number of down-up permutations on $\{1, 2, \dots, n-1, n+1, \dots, k+1\}$. Moreover, since the first number in an up-down permutation is n , the second number in this permutation must be greater than n . In other words, the first number in the corresponding down-up permutation on $\{1, 2, \dots, n-1, n+1, \dots, k+1\}$ must be greater than n . Therefore,

$$\begin{aligned} E_{k+1}^n &= \#\{1, 2, \dots, n-1, n+1, \dots, k+1\}_{n+1} \\ &+ \#\{1, 2, \dots, n-1, n+1, \dots, k+1\}_{n+2} \\ &+ \dots + \\ &+ \#\{1, 2, \dots, n-1, n+1, \dots, k+1\}_{k+1} \end{aligned}$$

Consider a term $\#\{1, 2, \dots, n-1, n+1, \dots, k+1\}_i$ ($i > n$) in the sum. Since this set contains k elements and the permutations start with the $(i-1)^{th}$ element, this term can be rewritten as $\#\{1, 2, \dots, n-1, n, \dots, k\}_{i-1}$. This latter term, by definition (2) is denoted as E_k^{i-1} .

Therefore $E_{k+1}^n = E_k^n + \dots + E_k^k$.

Apply rule (4) to construct row $k+1$ from the triangle with k rows, we also obtain: $X_{k+1,n} = X_{k,n} + \dots + X_{k,k} = E_k^n + \dots + E_k^k$. Hence $X_{k+1,n} = E_{k+1}^n$.

Secondly, we will show that $X_{k+2,n} = E_{k+2}^n$ by the same procedure.

Since $k+2$ odd, by definition (3), $E_{k+2}^n = \#\{1, 2, \dots, k+2\}_n$ is the number of down-up permutations on $\{1, 2, \dots, k+2\}$ starting with n . Since the first place is fixed with n , E_{k+2}^n is also the number of up-down permutations on $\{1, 2, \dots, n-$

$1, n+1, \dots, k+2\}$. Moreover, since the first number in an down-up permutation is n , the second number in this permutation must be less than n . In other words, the first number in the corresponding up-down permutation on $\{1, 2, \dots, n-1, n+1, \dots, k+1\}$ must be less than n . Therefore,

$$\begin{aligned} E_{k+2}^n &= \#\{1, 2, \dots, n-1, n+1, \dots, k+1\}_1 \\ &+ \#\{1, 2, \dots, n-1, n+1, \dots, k+1\}_2 \\ &+ \dots + \\ &+ \#\{1, 2, \dots, n-1, n+1, \dots, k+2\}_{n-1} \end{aligned}$$

Consider a term $\#\{1, 2, \dots, n-1, n+1, \dots, k+1\}_i$ ($i < n$) in the sum. Since this set contains $k+1$ elements and the permutations start with the i^{th} element, this term can be rewritten as $\#\{1, 2, \dots, n-1, n, \dots, k+1\}_i$. This latter term, by definition (3) is denoted as E_{k+1}^i .

Therefore $E_{k+2}^n = E_{k+1}^1 + \dots + E_{k+1}^{n-1}$.

Apply rule (5) to construct row $k+2$ from the triangle with $k+1$ rows, we also obtain: $X_{k+2\ n} = X_{k+1\ 1} + \dots + X_{k+1\ n-1} = E_{k+1}^1 + \dots + E_{k+1}^{n-1}$. Hence $X_{k+2\ n} = E_{k+2}^n$.

Therefore, each entry of the triangle can always be interpreted in terms of the number of alternating permutations. When k is odd (even), entries from left to right in row k of the triangle represent the numbers of down-up (up-down) permutations on a set of k elements starting from k to 1 respectively. The hypothesis has been proved. $\square$

The fact that we can interpret the triangle in terms of alternating permutations clarifies the meaning of non-zero entries at both sides of the triangle. When k is odd, the sum of k entries at row k is the number of down-up permutations on a set of k elements. This sum is also calculated by E_{k+1}^1 which is the entry at the right end of row $k+1$ since $E_{k+1}^1 = E_k^1 + \dots + E_k^k$. Similar argument regarding up-down permutations can be made for the entry E_{k+1}^{k+1} at the left end of row $k+1$ when k is even since $E_{k+1}^{k+1} = E_k^1 + \dots + E_k^k$.

Therefore, the entries at the ends of the triangle represent the numbers of up-down and down-up permutations alternatively on different sets of integers. However, the literature [Millar et. al., 2000; Arnold, 1991] seems to prefer interpreting entries at both sides of the triangle in terms of down-up permutations only. This may be done easily by a transform from the up-down permutation to the down-up permutation. For example, the up-down permutation 1 6 3 7 2 5 4 can be transformed into the down-up permutation 7 2 5 1 6 3 4 by subtracting each element in the up-down permutation from 8. In general, we can transform from an up-down permutation on $1, 2, \dots, k$ starting with n ($n \leq k$) to a down-up permutation on $1, 2, \dots, k$ starting with $k+1-n$. Thereby, when k is odd (even), entries from the left (right) to right (left) in row k of the triangle represent the number of down-up permutations on a set of k elements with their first places from k to 1 consequently.

6. THE TAYLOR SERIES OF $\tan x$ AND BERNOULLI NUMBERS

According to Wehrhahn [1992], the Bernoulli polynomials $\{B_n(x)\}$ are defined by the following properties:

$$\begin{aligned} B_0(x) &= 1 \\ B_n(x) &= \int n B_{n-1} dx \end{aligned}$$

$$B_n(x) = \int_0^1 B_n(x) dx = 0, \text{ for } n \geq 1$$

The n^{th} Bernoulli number B_n is defined by:

$$B_n = B_n(0), \text{ for } n = 0, 1, 2, \dots$$

The first few terms from the Bernoulli sequence are shown in the following table:

n	0	1	2	3	4	5	6	7	8	9	10
B_n	1	$-\frac{1}{2}$	$\frac{1}{6}$	0	$-\frac{1}{30}$	0	$\frac{1}{42}$	0	$-\frac{1}{30}$	0	$\frac{5}{66}$

These Bernoulli numbers have the following exponential generating function:

$$\frac{t}{e^t - 1} = \sum_{n=0}^{\infty} B_n \frac{t^n}{n!}, \quad (|t| < 2\pi)$$

It is also well known that the coefficient $B_{2k+1} = 0$ holds for $k > 0$ since:

$$\begin{aligned} \frac{t}{e^t - 1} &= B_0 \frac{t^0}{0!} + B_1 \frac{t^1}{1!} + \sum_{k=2}^{\infty} B_k \frac{t^k}{k!} = 1 - \frac{t}{2} + \sum_{k=2}^{\infty} B_k \frac{t^k}{k!} \\ \Rightarrow \frac{t}{e^t - 1} + \frac{t}{2} &= 1 + \sum_{k=2}^{\infty} B_k \frac{t^k}{k!} \end{aligned}$$

The left component:

$$\frac{t}{e^t - 1} + \frac{t}{2} = \frac{t}{2} \left(\frac{2}{e^t - 1} + 1 \right) = \frac{t}{2} \frac{e^t + 1}{e^t - 1} = \frac{t}{2} \frac{e^{\frac{t}{2}} + e^{-\frac{t}{2}}}{e^{\frac{t}{2}} - e^{-\frac{t}{2}}} = \frac{t}{2} \coth \frac{t}{2}$$

is an even function, which implies that the coefficients corresponding to the odd powers in the Taylor expansion are 0. Hence $B_{2k+1} = 0$ holds for $k > 0$.

The relationship between the coefficients of the Taylor expansion of $\tan x$ and the Bernoulli numbers can be proved as follows:

$$\begin{aligned} e^{ix} &= 1 + \frac{ix}{1!} + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \dots + \frac{(ix)^n}{n!} + \dots \\ e^{-ix} &= 1 + \frac{-ix}{1!} + \frac{(-ix)^2}{2!} + \frac{(-ix)^3}{3!} + \frac{(-ix)^4}{4!} + \dots + \frac{(-ix)^n}{n!} + \dots \\ \frac{e^{ix} + e^{-ix}}{2} &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots + (-1)^n \frac{x^{2n}}{(2n)!} + \dots = \cos x \\ \text{and } \frac{e^{ix} - e^{-ix}}{2i} &= x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \dots = \sin x \end{aligned}$$

Therefore,

$$\begin{aligned} \tan x &= \frac{\sin x}{\cos x} = \frac{e^{ix} - e^{-ix}}{2i} \frac{2}{e^{ix} + e^{-ix}} = \frac{e^{ix} - e^{-ix}}{e^{ix} + e^{-ix}} \frac{1}{i} = \frac{e^{2ix} - 1}{e^{2ix} + 1} (-i) \\ &= \frac{e^{4ix} - 2e^{2ix} + 1}{e^{4ix} - 1} (-i) = \frac{(e^{4ix} - 1) - 2(e^{2ix} + 1) + 4}{e^{4ix} - 1} (-i) \\ &= \left(1 - 2 \frac{1}{e^{2ix} - 1} + \frac{4}{e^{4ix} - 1} \right) (-i) = \frac{2i}{e^{2ix} - 1} - \frac{4}{e^{4ix} - 1} - i \end{aligned}$$

Since the exponential generating function for the Bernoulli numbers is

$$\frac{t}{e^t - 1} = \sum_{n=0}^{\infty} B_n \frac{t^n}{n!}, \quad (|t| < 2\pi)$$

we have:

$$\begin{aligned}
x \tan x &= \frac{2ix}{e^{2ix} - 1} - \frac{4ix}{e^{4ix} - 1} - ix \\
&= \sum_{n=0}^{\infty} B_n \frac{(2ix)^n}{n!} - \sum_{n=0}^{\infty} B_n \frac{(4ix)^n}{n!} - ix \\
&= \left(1 + B_1 \cdot (2ix) + \sum_{n=2}^{\infty} B_n \frac{(2ix)^n}{n!} \right) \\
&\quad - \left(1 + B_1 \cdot (4ix) + \sum_{n=2}^{\infty} B_n \frac{(4ix)^n}{n!} \right) - ix \\
&= \sum_{n=2}^{\infty} B_n \frac{(2ix)^n}{n!} - \sum_{n=2}^{\infty} B_n \frac{(4ix)^n}{n!} \\
&= \sum_{n=2}^{\infty} B_n \frac{2^n (ix)^n}{n!} - \sum_{n=2}^{\infty} B_n \frac{2^{2n} (ix)^n}{n!} \\
&= \sum_{n=2}^{\infty} (2^n - 2^{2n}) B_n \frac{(ix)^n}{n!}
\end{aligned}$$

Since $B_{2k+1} = 0$ for $k \geq 2$, only B_{2k} is involved in the Taylor series:

$$\begin{aligned}
x \tan x &= \sum_{k=1}^{\infty} (2^{2k} - 2^{2 \cdot 2k}) B_{2k} i^{2k} \frac{x^{2k}}{(2k)!} \\
&= \sum_{k=1}^{\infty} \frac{2^{2k} (1 - 2^{2k}) (-1)^k}{2k} B_{2k} \frac{x^{2k}}{(2k-1)!} \\
&= \sum_{k=1}^{\infty} \frac{2^{2k} (2^{2k-1}) (-1)^{k+1}}{2k} B_{2k} \frac{x^{2k}}{(2k-1)!}
\end{aligned}$$

Convergence conditions:

$$\left. \begin{aligned} |2ix| < 2\pi &\Rightarrow |4i^2 x^2| < 4\pi^2 \Rightarrow 4x^2 < 4\pi^2 \Rightarrow |x| < \pi \\ |4ix| < 2\pi &\Rightarrow |16i^2 x^2| < 4\pi^2 \Rightarrow 16x^2 < 4\pi^2 \Rightarrow |x| < \frac{\pi}{2} \end{aligned} \right\} \Rightarrow |x| < \frac{\pi}{2}$$

The Taylor series for $\tan x$ is:

$$\tan x = \sum_{k=1}^{\infty} T_k \frac{x^{2k-1}}{(2k-1)!} \Rightarrow x \tan x = \sum_{k=1}^{\infty} T_k \frac{x^{2k}}{(2k-1)!}$$

Equating the two Taylor series, we have:

$$T_k = \frac{(-1)^{k+1} 2^{2k} (2^{2k} - 1)}{2k} B_{2k}, \quad \text{for } k \neq 0$$

This formula shows that the coefficients of the Taylor expansion of $\tan x$ are proportional to the Bernoulli numbers. Since $B_{2k+1} = 0$ for $k > 0$, the literature sometimes refer B_k to the k^{th} non-zero values of the Bernoulli numbers. Moreover, a Bernoulli number B_k may also be considered as a solely positive number by taking the values of $(-1)^{k+1} B_k$. These notations bring us to a more concise formula:

$$T_k = \frac{4^k (4^k - 1)}{2k} B_{k+1}$$

In conclusion, the report focuses on the relationship among completely ordered leveled binary trees, the Pascal-like triangle and the coefficients of the Taylor expansion of $\tan x$ and $\sec x$. These relationships have been established through their individual connections with the alternating permutations by the recursion method.

Additionally, some examination about the relationship between the Bernoulli numbers and the Taylor series of $\tan x$ has also been presented since our investigation arose from the COL binary trees which are highly correlated with the latter. However, the symmetry of $\tan x$ and $\sec x$ may suggest further research on the Taylor expansion of $\sec x$.

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APPENDIX A.

An example of row 5 represents down-up permutations on the set $\{1, 2, 3, 4, 5\}$:

There is 0 down-up permutations starting with 1.

There are 2 down-up permutations starting with 2:

21435

21534

There are 4 down-up permutations starting with 3:

31425

31524

32415

32514

There are 5 down-up permutations starting with 4:

41325

41523

42315

42513

43512

There are 5 down-up permutations starting with 5:

51324

51423

52314

52413

53412

The entry 16 at the right end of row 6 is the sum of 0, 2, 4, 5, 5. Hence it is the total number of down-up permutations on the set $\{1, 2, 3, 4, 5\}$.

APPENDIX B.

The Taylor expansion of trigonometric functions are given as follows:

$$\begin{aligned}\sin x &= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} \\ \cos x &= \sum_{m=0}^{\infty} (-1)^m \frac{x^{2m}}{(2m)!} \\ \sec x &= \sum_{n=0}^{\infty} E_n \frac{x^{2n}}{(2n)!} \\ \tan x &= \sum_{n=0}^{\infty} T_n \frac{x^{2n+1}}{(2n+1)!}\end{aligned}$$

The first recursive formula can be proved as follows:

$$\begin{aligned}&\overset{\sec x \cos x = 1}{\Rightarrow} \left(\sum_{n=0}^{\infty} E_n \frac{x^{2n}}{(2n)!} \right) \left(\sum_{m=0}^{\infty} (-1)^m \frac{x^{2m}}{(2m)!} \right) = 1 \\ &\Rightarrow \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} (-1)^m E_n \frac{x^{2m+2n}}{(2m)!(2n)!} = 1\end{aligned}$$

Let $k = m + n \Rightarrow n \leq k$, we have:

$$\sum_{k=0}^{\infty} \sum_{n=0}^k (-1)^{k-n} E_n \frac{x^{2k}}{(2k-2n)!(2n)!} = 1$$

$$\begin{aligned}
&\Rightarrow \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!} \sum_{n=0}^k (-1)^{k-n} E_n \frac{(2k)!}{(2k-2n)!(2n)!} = 1 \\
&\Rightarrow \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!} \sum_{n=0}^k (-1)^{k-n} \binom{2k}{2n} E_n = 1 \\
&\Rightarrow \sum_{n=0}^k (-1)^{k-n} \binom{2k}{2n} E_n = 0 \\
&\Rightarrow \sum_{n=0}^k (-1)^{n-k} \binom{2k}{2n} E_n = 0 \\
&\Rightarrow (-1)^k \sum_{n=0}^k (-1)^{n-k} \binom{2k}{2n} E_n = 0 \\
&\Rightarrow \sum_{n=0}^k (-1)^n \binom{2k}{2n} E_n = 0
\end{aligned}$$

The second recursive formula can be proved as follows:

$$\begin{aligned}
&\Rightarrow \left(\sum_{m=0}^{\infty} (-1)^m \frac{x^{2m}}{(2m)!} \right) \left(\sum_{n=0}^{\infty} T_n \frac{x^{2n+1}}{(2n+1)!} \right) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} \\
&\Rightarrow \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} (-1)^m T_n \frac{x^{2m+2n+1}}{(2m)!(2n)!} = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}
\end{aligned}$$

Let $k = m + n \Rightarrow n \leq k$, we have:

$$\begin{aligned}
&\sum_{k=0}^{\infty} \sum_{n=0}^k (-1)^{k-n} T_n \frac{x^{2k+1}}{(2k-2n)!(2n+1)!} = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} \\
&\Rightarrow \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!} \sum_{n=0}^k (-1)^{k-n} T_n \frac{(2k+1)!}{(2k-2n)!(2n+1)!} = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} \\
&\Rightarrow \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!} \sum_{n=0}^k (-1)^{k-n} T_n \binom{2k+1}{2n+1} = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} \\
&\Rightarrow \sum_{n=0}^k (-1)^{k-n} \binom{2k+1}{2n+1} T_n = (-1)^k \\
&\Rightarrow \sum_{n=0}^k (-1)^{-n} \binom{2k+1}{2n+1} T_n = 1 \\
&\Rightarrow \sum_{n=0}^k (-1)^n \binom{2k+1}{2n+1} T_n = 1
\end{aligned}$$

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